

Noise-Robust PINN-Based Admittance Identification for Grid-Connected Voltage Source Converters

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Abstract

This report documents the full body of work carried out during a research internship on machine-learning-based online admittance identification of grid-connected Voltage Source Converters (VSCs). The project began as a reproduction study of the physics-informed neural network (PINN) method of Zhang, Xu, and Wang (IEEE Trans. Industrial Electronics, 2023), which identifies the 2×2 dq-frame admittance matrix of a VSC from limited online measurements. The analytical small-signal model, the restructured neural-network architecture, and the paper’s core case studies were reproduced from first principles, since no official code release accompanies the publication. A critical assessment of the reproduced method concluded that its “physics-informed” compression functions largely as a substitute for standard frequency-domain extraction (FFT-based) tools, rather than as a fundamentally new identification capability. This finding, discussed with the supervisor, motivated a pivot toward an independent noise-robust, time-domain system identification approach. Two complementary pipelines were subsequently developed and implemented in Python/PyTorch with MATLAB-generated training data:

- (i) *Gray-box* parameter estimator: regresses physical circuit and controller parameters directly from noisy step-response data, refined with a physics-informed fine-tuning stage.
- (ii) *Black-box* transfer-function identifier: predicts stable pole-zero parametrizations of each admittance component independently, evaluated across a range of signal-to-noise ratios and noise statistics.

This report presents the theoretical background, the reproduction results, the rationale for the pivot, the full methodology of the proposed approach, the implementation details of both pipelines, and the experimental results obtained to date, followed by a discussion, conclusions.

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Chapter 1

Introduction

1.1 Motivation

The increasing penetration of power-electronic-interfaced renewable generation is changing the fundamental character of modern power systems. Voltage Source Converters (VSCs) are the universal interface between renewable sources – wind, solar, battery storage – and the AC grid, and their proliferation has introduced a class of stability problems that classical synchronous-generator-based analysis does not capture: converter–grid interaction instability, driven by the closed-loop dynamics of the converter’s control system rather than by electromechanical swing dynamics. Impedance-based (or admittance-based) stability analysis has become the standard practical framework for studying this phenomenon, because it allows the grid and the converter to be characterized independently, at their point of common coupling (PCC), and their interaction assessed using frequency-domain criteria such as the Generalized Nyquist Criterion (GNC), without requiring a full internal model of either subsystem.

The practical difficulty is that converter manufacturers do not disclose control-loop structures or parameters. The converter impedance/admittance must therefore be identified as a black box, typically by injecting a perturbation at the PCC and measuring the resulting voltage/current response. Two broad families of identification approach exist:

- **Frequency-domain / harmonic injection methods**, which sweep or inject sinusoidal perturbations and extract the admittance via FFT-based frequency-response estimation. These are accurate but require dedicated hardware, repeated excitations, and are slow – making them poorly suited to *online*, continuously changing operating conditions.
- **Time-domain system identification methods**, such as the Eigensystem Realization Algorithm (ERA) or Dynamic Mode Decomposition (DMD), which extract a state-space or transfer-function model directly from transient response data. These are fast and require only short measurement windows, but are highly sensitive to measurement noise, since their core step (an SVD of a Hankel-structured data matrix) directly propagates noise into the identified model order and residues.

Machine learning offers a third path: a trained model can, in principle, learn a noise-robust mapping from raw time-domain measurements to admittance information, without requiring repeated excitations and without the fragility of a purely linear-algebraic decomposition of noisy data. This is the premise investigated in this internship, beginning with a critical reproduction of a recent representative method in this space.

1.2 Objectives

The internship had two sequential objectives, corresponding to its two phases:

1. **Phase 1 – Reproduction.** Reproduce, from first principles and without an accompanying code release, the physics-informed neural network (PINN) based online impedance identification method of Zhang et al. (2023), including its analytical derivation, its restructured neural-network architecture, and its case studies, in order to critically evaluate the actual contribution of the method.
2. **Phase 2 – Original methodology.** Based on the conclusions of Phase 1, design, implement, and evaluate an alternative, noise-robust, time-domain admittance identification method that does not rely on frequency-domain extraction tools, does not constrain the network architecture to a specific converter’s physics, and generalizes to unknown control structures.

1.3 Report Structure

Chapter 2 introduces the power-systems and control-theory background needed to follow the remainder of the report. Chapter 3 details the reproduction of Zhang et al. (2023): the analytical model, the restructured ANN, the two-step transfer-learning procedure, and the reproduced case studies. Chapter 4 presents the critical assessment of the reproduced method and the reasoning that led to the change of research direction. Chapter 5 describes the proposed noise-robust identification methodology in full. Chapters 6 and 7 describe, respectively, the gray-box parameter-estimation pipeline and the black-box transfer-function pipeline that were implemented and evaluated. Chapter 8 presents the experimental results obtained across both pipelines. Chapter 9 discusses these results, and Chapter 10 concludes the report .

Chapter 2

Background

2.1 The Converter–Grid Interaction Problem

A grid-connected VSC regulates its output current (or power) through a cascaded control structure: an inner current-control loop, typically implemented in the synchronously-rotating dq reference frame with proportional-integral (PI) or proportional-resonant (PR) controllers, and an outer synchronization loop – the phase-locked loop (PLL) – that tracks the grid voltage angle. The converter is connected to the grid through an output filter (L, LC, or LCL) and, electrically, appears at its terminal as a nonlinear, time-varying, but locally linearizable two-port network. Around a given steady-state operating point, its small-signal behaviour is fully captured by a 2×2 complex transfer-function matrix relating small perturbations in dq-frame terminal voltage to the resulting dq-frame current:

$$\begin{bmatrix} \Delta I_d(s) \\ \Delta I_q(s) \end{bmatrix} = - \begin{bmatrix} Y_{dd}(s) & Y_{dq}(s) \\ Y_{qd}(s) & Y_{qq}(s) \end{bmatrix} \begin{bmatrix} \Delta V_d(s) \\ \Delta V_q(s) \end{bmatrix}. \quad (2.1)$$

This matrix, $Y_{dq}(s)$, is the converter’s *dq-frame admittance*. Because the PLL couples the d- and q-axis dynamics, the off-diagonal terms Y_{dq}, Y_{qd} are, in general, non-zero, unlike in a simple decoupled circuit.

Once $Y_{dq}(s)$ is known at the point of common coupling, together with the grid’s own Thevenin admittance $Y_g(s)$, the stability of the interconnected system can be assessed via the Generalized Nyquist Criterion applied to the loop-gain matrix $Y_{dq}(s)Z_g(s)$ (or the equivalent minor-loop gain formulation), without needing a joint state-space model of converter and grid together. This is the practical payoff that motivates admittance identification: a correctly identified $Y_{dq}(s)$, refreshed online as the operating point drifts, enables continuous stability monitoring of a system whose internal converter structure is unknown.

2.2 Small-Signal Modeling of a Grid-Connected VSC

For a standard current-controlled VSC with an L-type output filter, PI current control, and a synchronous-reference-frame PLL, the small-signal admittance can be derived analytically from first principles. This derivation, followed during the reproduction phase, proceeds through the following steps.

2.2.1 Filter and Plant Dynamics

The converter output filter (inductance L_f , parasitic resistance R_f) obeys, in the dq frame rotating at the grid fundamental frequency ω_1 , the coupled differential equations

$$L_f \frac{d\Delta i_d}{dt} = \Delta v_{o,d} - \Delta v_d - R_f \Delta i_d + \omega_1 L_f \Delta i_q, \quad (2.2)$$

$$L_f \frac{d\Delta i_q}{dt} = \Delta v_{o,q} - \Delta v_q - R_f \Delta i_q - \omega_1 L_f \Delta i_d, \quad (2.3)$$

where v_o is the converter-side voltage and v is the PCC (grid-side) voltage. These two equations, together with the states of the current controller and the PLL, form a linear time-invariant (LTI) small-signal state-space system

$$\dot{x} = Ax + Bu, \quad y = Cx + Du, \quad (2.4)$$

whose A, B, C, D matrices are parametrized directly by the physical and control parameters: filter inductance L_f and resistance R_f , current-loop PI gains $K_{p,i}, K_{i,i}$, PLL PI gains $K_{p,pll}, K_{i,pll}$, and the steady-state operating point $(I_{d,eq}, I_{q,eq}, V_{dc})$. This mapping – physical/control parameters $\rightarrow$ state-space matrices $\rightarrow$ admittance $Y_{dq}(s)$ – is implemented as the function `build_ABCD` in the project codebase and forms the analytical backbone of both the reproduction and the subsequent original work.

2.2.2 PLL Linearization

The synchronous-reference-frame PLL introduces a feedback path between the measured grid-side voltage angle error and the reference frame used for the dq transformation. Linearizing its dynamics around the operating point yields a scalar transfer function

$$G_{PLL}(s) = \frac{s K_{p,pll} + K_{i,pll}}{s^2 + (s K_{p,pll} + K_{i,pll}) V_d}, \quad (2.5)$$

which couples into the current-loop dynamics through the dq-transformation of both the measured current and the measured voltage. The resulting cross-coupling terms are what make the off-diagonal admittance components Y_{dq}, Y_{qd} non-zero and operating-point dependent.

2.2.3 Closed-Loop Admittance

Combining the filter dynamics, the current controller, the computational and PWM time delay $G_{del}(s) = e^{-1.5T_s s}$, and the PLL coupling terms yields the closed-loop admittance matrix

$$Y_{dq}(s) = (I + T_{dq}(s))^{-1} Y_{to,dq}(s) - G_{cl,dq}(s) Y_{PLL}(s), \quad (2.6)$$

where $T_{dq}(s)$ is the current-loop gain matrix, $Y_{to,dq}(s)$ is the open-loop filter admittance modified by the delay and PLL feed-through, and $G_{cl,dq}(s)$ is the closed-loop current reference-to-output transfer matrix. Equation (2.6) makes explicit that the admittance depends jointly on frequency *and* on the steady-state operating point (through $I_{d,eq}, I_{q,eq}, V_d, V_q$), which is the origin of the *multi-operating-point* (MOP) admittance model concept used throughout this work.

2.3 Machine-Learning Approaches to Admittance Identification

Three broad ML-adjacent strategies for admittance/impedance identification were considered during this project, and are contrasted below; the first was the subject of the reproduction phase, the second was investigated and rejected, and the third is the approach ultimately adopted.

Physics-informed neural network (PINN), architecture-embedded physics.

The network's layer structure is derived directly from the analytical admittance expression (Eq. (2.6)), so that the trained weights correspond to physically meaningful sub-functions. This compresses the hypothesis space and reduces the data required for training, but ties the network's applicability to converters whose control structure matches the assumed topology.

Time-domain linear system identification (ERA/DMD).

A Hankel matrix is built from transient step-response data and factorized via SVD to recover a minimal-order state-space realization. This requires no training and no assumed control structure, but its core step is a direct SVD of noisy data, so its accuracy degrades sharply as measurement SNR decreases.

Noise-robust ANN with physics confined to label generation (adopted approach).

A generic feedforward network is trained, offline, on a large synthetic population of random stable linear systems (not VSC-specific), to map noisy time-domain signals to admittance information. Physics is used only to generate correct training labels (via modal decomposition) and to constrain output parametrizations to stable systems – not to shape the network architecture itself. This decouples the network's applicability from any specific converter topology.

Chapter 3

Reproduction of the PINN-Based Method (Zhang et al., 2023)

3.1 Overview of the Reproduced Paper

The reproduced publication is M. Zhang, Q. Xu, and X. Wang, “Physics-Informed Neural Network Based Online Impedance Identification of Voltage Source Converters,” *IEEE Transactions on Industrial Electronics*. No official code implementation accompanies the paper, so the entire pipeline – analytical model, network architecture, training procedure, and case studies – was reconstructed from the text and figures of the publication itself, following an internally-authored, phased reproduction guide.

The paper’s central claim is that partial physics knowledge of a general VSC can be used to *restructure* (rather than merely initialize or regularize) an artificial neural network, so that:

1. the required training data volume is significantly reduced relative to a randomly-initialized ANN of comparable capacity, and
2. an *offline* model trained on a known “general” VSC can be efficiently transferred, via a small amount of *online* field-measured data, to identify an *unknown* target VSC (including one with a different control structure), using transfer-learning theory.

3.2 Reproduced Analytical Model

The full small-signal derivation of Section 2.2 was implemented in MATLAB, including the filter dynamics, PI current controller, SRF-PLL linearization, digital control delay, and the resulting closed-loop admittance matrix of Eq. (2.6). This was validated by comparison against Bode-plot sweeps of the equivalent Simulink/Simscape Electrical switching model, confirming that the linearized analytical model matches the small-signal behaviour of the full nonlinear simulation around each tested operating point.

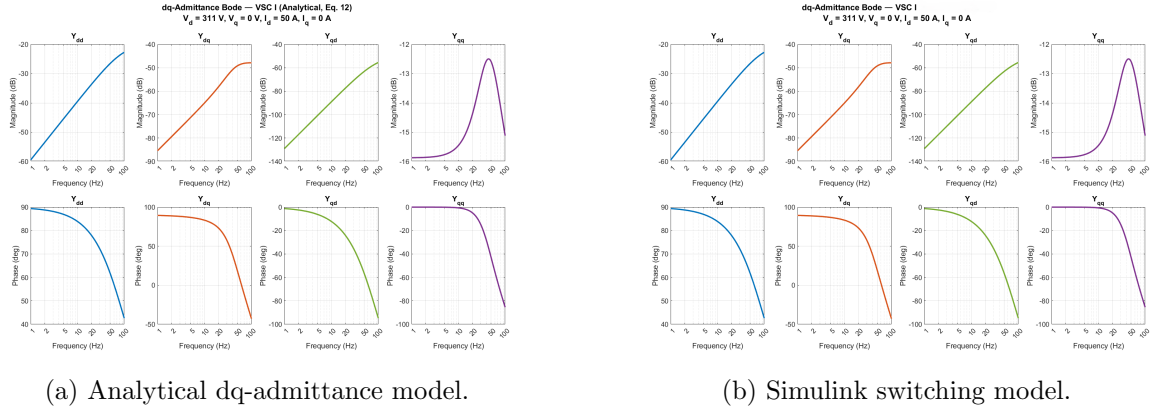


Figure 3.1: Bode-plot validation of the analytical dq-admittance model against the Simulink switching model for VSC I parameters, $I_d = 50$ A, comparing Y_{dd} , Y_{dq} , Y_{qd} , and Y_{qq} .

The two panels of Fig. 3.1 overlay closely across the full 1–100 Hz band for all four admittance components, in both magnitude and phase, confirming that the linearized closed-loop model of Eq. (2.6) correctly captures the small-signal dynamics of the switching-level Simulink model at this operating point. The direct-axis term Y_{dd} shows the expected inductive roll-off with phase approaching $+90^\circ$ at low frequency, while Y_{qq} exhibits the PLL-induced resonant feature discussed in Section 2.2; both features are reproduced by the switching model, which is the strongest available evidence that the analytical derivation and its MATLAB implementation are correct before any learning-based stage is introduced.

3.3 Dataset Inspection: Dense and Sparse Admittance Surfaces

Before any network is trained, the notebook performs a direct visual inspection of the raw admittance datasets used later for training and validation, reproducing the paper’s Figs. 9 and 11. Two dedicated routines, `plot_raw_dense_data` and `plot_raw_sparse_data`, render the magnitude and phase of each admittance component as 3-D surfaces over the swept frequency and operating-current range, using axis limits matched to the paper (`FIG9_LIMITS`, `FIG11_LIMITS`) so that the reproduced surfaces can be compared directly against the published figures by eye.

3.3.1 Restructured Neural Network Architecture

Following the paper’s Section III-A2, the closed-loop admittance expression was algebraically decomposed into an operating-point-dependent part and a frequency-dependent part, matching the paper’s Eq. (13)–(18). This decomposition allows the admittance to be written, per axis component, as

$$Y_{dd}(s) = G[V_q, V_d, I_q, I_d, F_1(s), F_2(s), F_3(s), F_4(s), F_5(s)], \quad (3.1)$$

i.e., as a known algebraic function G (a weighted sum) of the steady-state operating point and of five frequency-dependent sub-functions $F_1(s), \dots, F_5(s)$. This structure was mirrored directly into a three-layer network, implemented as three explicit PyTorch modules:

- **Layer I** (**LayerIFrequencyBranch**) – a SiLU-activated MLP (3 hidden layers, 64 units) taking a 4-dimensional frequency feature vector as input and producing 10 outputs: the real and imaginary parts of the five sub-functions $F_1(s), \dots, F_5(s)$;
- **Layer II** (**LayerIIOperatingCombiner**) – a fixed, non-trainable linear combination implementing $Y_{\text{real}} = F_{1,r} - V_q F_{2,r} - V_d F_{3,r} - I_q F_{4,r} - I_d F_{5,r}$ (and equivalently for the imaginary part), taking the Layer-I outputs and the steady-state operating point (V_d, V_q, I_d, I_q) as inputs;
- **Layer III** (**LayerIIIDecoder**) – magnitude/phase extraction, $\text{mag}_{\text{dB}} = 20 \log_{10}|Y|$ and $\text{phase} = \text{atan2}(Y_{\text{imag}}, Y_{\text{real}})$.

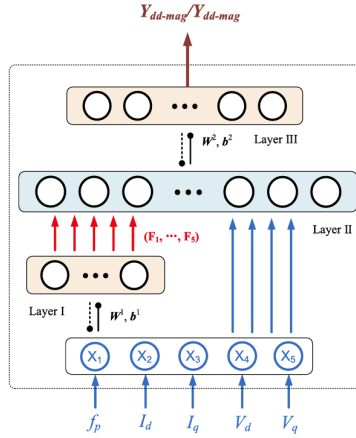


Figure 3.2: Reconstructed network architecture for admittance-component identification, adapted from the reference paper.

These three modules are composed into a single **FullPINNSystem**, trained independently for each of the four admittance components (Y_{dd} , Y_{dq} , Y_{qd} , Y_{qq}), following the architecture described in the paper’s Fig. 5.

3.4 Conventional-ANN Baseline: Two Variants

Before evaluating the PINN, the reproduction establishes a conventional (non-physics-restructured) ANN baseline against which the PINN’s data efficiency is measured. Two architecturally distinct variants were implemented and benchmarked against each other, following the paper’s Fig. 14 comparison:

- **Variant A – Unified network.** A single **UnifiedConventionalANN** maps the 5-dimensional input (f, V_d, V_q, I_d, I_q) directly to all 8 outputs (real and imaginary parts of $Y_{dd}, Y_{dq}, Y_{qd}, Y_{qq}$) simultaneously, through a shared 64-unit, 2-hidden-layer MLP trained for 2000 epochs on the sparse VSC II dataset.
- **Variant B – Element-specific networks.** Four independent **ElementSpecificANN** instances are trained separately, one per admittance component, on the same sparse dataset.

Both variants are evaluated on the same dense held-out VSC II grid, and their magnitude/phase error surfaces are rendered following the paper’s Fig. 14 layout.

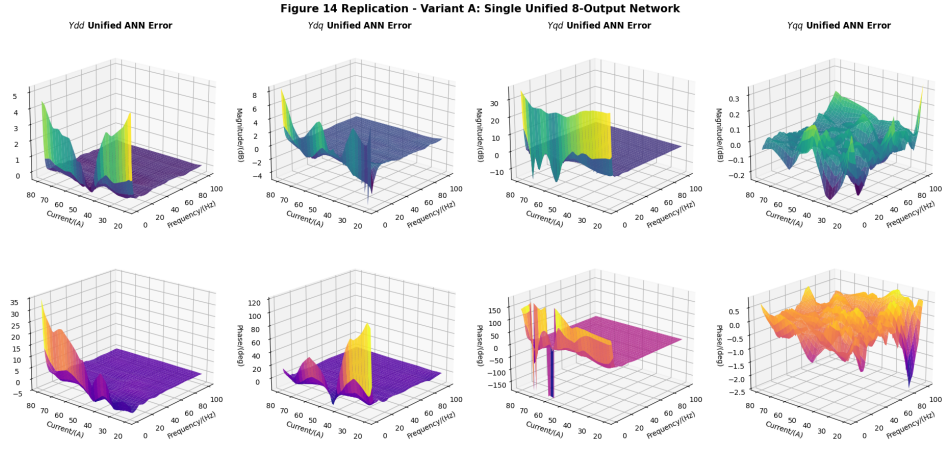


Figure 3.3: Variant A (unified 8-output network) magnitude and phase error surfaces against the dense VSC II validation grid, reproducing the paper's Fig. 14 layout.

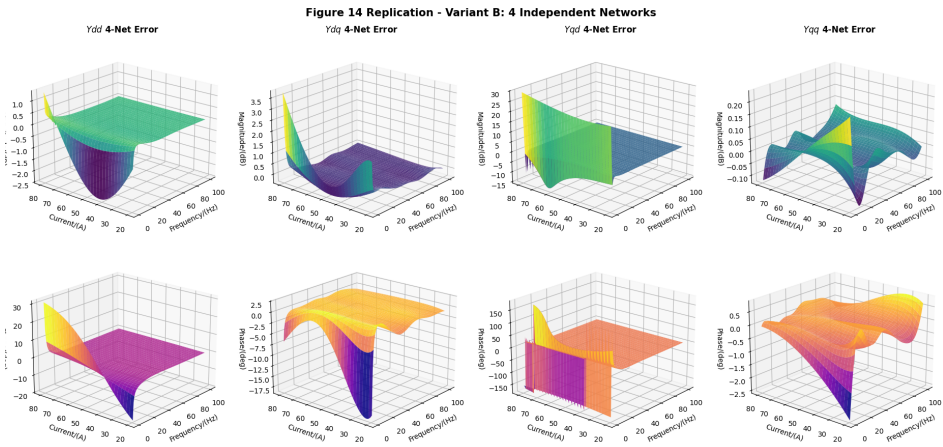


Figure 3.4: Variant B (four independent element-specific networks) magnitude and phase error surfaces against the dense VSC II validation grid.

Establishing which conventional-ANN variant is the stronger baseline before comparing either to the PINN is important for the conclusion of Chapter 4: the PINN’s data-efficiency advantage must be measured against the *better* of the two conventional baselines, not against whichever is arbitrarily weaker, or the resulting comparison overstates the benefit of architectural physics restructuring.

3.5 Offline/Online Two-Step Identification Procedure

The paper’s transfer-learning procedure was reproduced in two stages:

1. **Offline phase.** A “known” converter, VSC I, is simulated over a dense grid of operating points (I_d from 20 to 70 A, frequency f_p from 1 to 100 Hz), and the restructured **FullPINNSystem** is trained on this dense synthetic dataset, independently per component, to produce an offline multi-operating-point (MOP) admittance model $Y_{\text{VSC-I}}$ (paper Fig. 10).
2. **Online phase.** A “target” converter, VSC II – treated as a black box, with a different set of physical/control parameters (and, in the second case study, a different filter topology with active damping) – is measured at only a sparse set of operating points (200 points, versus thousands in the offline phase). This sparse dataset is used to fine-tune the offline model, producing the online model $Y_{\text{VSC-II}}$ (paper Figs. 12–13).

3.5.1 Three-Tier Automated Transfer Strategy

A detail not made explicit in the paper’s description of Section III-B, but required to obtain a reliably convergent fine-tuning stage during reproduction, is an automated, per-component, three-tier fallback strategy for the online transfer step:

1. **Tier 1 – Simple fine-tuning** (`finetune_simple`): all Layer-I weights of the offline-trained model are fine-tuned directly on the sparse online dataset at a learning rate of 3×10^{-4} .
2. **Tier 2 – Gradual unfreeze** (`finetune_gradual_unfreeze`): if Tier 1 fails to reach the target validation loss, a fraction (30%) of the network is kept frozen at a higher learning rate (10^{-3}) while the remainder is fine-tuned at a lower rate (10^{-4}), progressively unfreezing layers.
3. **Tier 3 – Transfer Component Analysis (TCA) subspace alignment** (`compute_tca_alignment`): if Tiers 1–2 both fail, a domain-adaptation step aligns the offline (source) and online (target) feature distributions in a shared 10-component subspace, following the transfer-learning formulation cited by the paper (Pan and Yang, 2010; Pan et al., 2011).

Each component ($Y_{dd}, Y_{dq}, Y_{qd}, Y_{qq}$) escalates through these tiers independently, governed by a validation-loss threshold of 5×10^{-4} (corresponding to approximately 1% identification error), with component-specific epoch budgets and restart counts (Table 3.1).

Table 3.1: Per-component transfer configuration for the three-tier fine-tuning strategy.

Component	Fine-tuning epochs	Restarts
Y_{dd}	800	2
Y_{dq}	1500	5
Y_{qd}	1500	5
Y_{qq}	800	2

The larger epoch/restart budget allocated to the cross-coupling terms Y_{dq}, Y_{qd} reflects their smaller magnitude and greater sensitivity to the PLL-coupling parameters, making them harder to fit reliably than the direct terms Y_{dd}, Y_{qq} .

3.6 Reproduced Case Studies

3.6.1 Case Study – PI-Controlled VSC (VSC I $\rightarrow$ VSC II)

Following Table I of the paper, VSC I ($f_{sw} = 20$ kHz, $L_f = 1$ mH, $R_f = 0.3$ m Ω , $K_{p.i} = 10.5$, $K_{i.i} = 5900$) was used to generate the dense offline training dataset (Fig. 9 of the paper), and the resulting offline MOP-admittance model (Fig. 10) was transferred, using only 200 sparse online points from VSC II ($f_{sw} = 10$ kHz, $L_f = 3$ mH, $R_f = 0.9$ m Ω , $K_{p.i} = 31.4$, $K_{i.i} = 16449$), to produce the online admittance model of VSC II (Figs. 11–12). The identification error against dense field-measured validation data (Fig. 13) and against a conventional, randomly-initialized ANN trained on the same sparse dataset (Fig. 14) were reproduced.

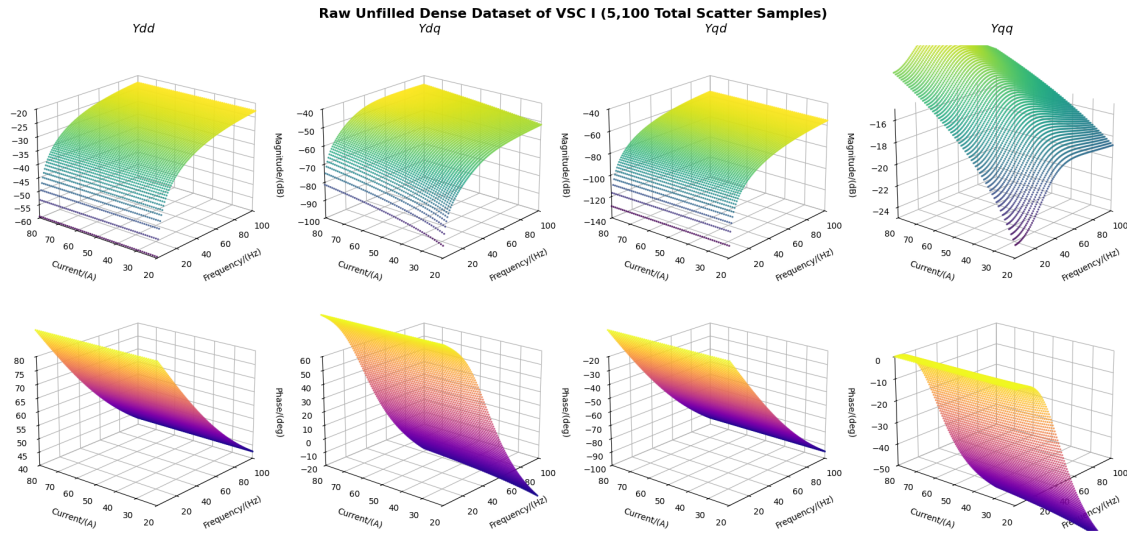


Figure 3.5: Offline PINN admittance model trained on the dense VSC I dataset, reproducing the paper’s Fig. 10.

Figure 3.5 shows the offline MOP-admittance surfaces generated by the restructured network once trained on the dense VSC I dataset, evaluated across the full swept operating range. The magnitude surfaces closely track the shape of the analytical training data across the full I_d and frequency range, including the Y_{qq} resonant feature, indicating that the offline training stage has converged to a model that generalizes smoothly across the operating-point grid rather than merely interpolating between training samples.

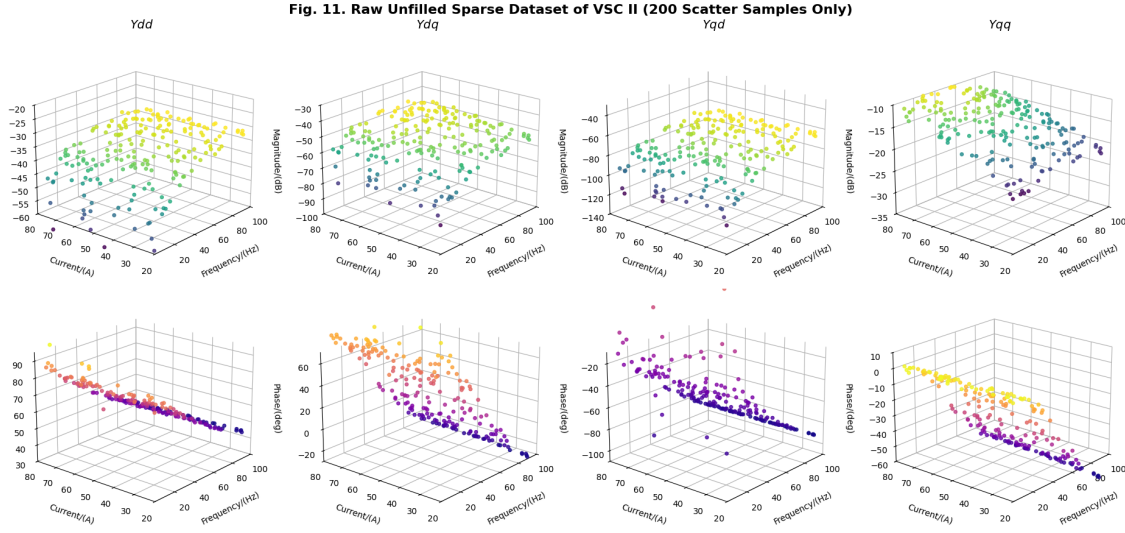


Figure 3.6: Reproduced generated MOP-admittance model of VSC II (cf. paper Fig. 11).

Figure 5.1 is the corresponding admittance model for the target converter VSC II, obtained after the three-tier online transfer step of Section 3 is applied to only 200 sparse field-measured points. The overall shape of the surfaces is preserved relative to VSC I, but with the magnitude and resonant-frequency shifts expected from VSC II's different filter inductance and controller gains (Table I), confirming that the transfer step adapts the offline model to the new converter's parameters rather than simply reproducing the source model unchanged.

Chapter 4

Critical Assessment and Rationale for the Research Pivot

4.1 What the Reproduction Revealed

Having implemented the full pipeline of Zhang et al. (2023) end to end, its actual mechanism of benefit can be stated precisely: the restructured network’s Layer I sub-networks ($F_1(s), \dots, F_5(s)$) are learning frequency-domain transfer functions that are already fully determined, in closed form, by the analytical small-signal model of Section 2.2 once the physical/control parameters are known – and even when they are not fully known, these frequency-response curves are exactly what conventional FFT-based frequency-sweep identification already extracts directly and non-parametrically from measured data, without any training at all. In other words, the “physics-informed compression” is functioning primarily as a *data-efficient replacement for an FFT*, rather than as a qualitatively new identification capability: it reduces the number of operating points that need to be densely sampled online, at the cost of requiring an offline-trained model of a *known*, structurally similar converter to transfer from.

4.2 Limitations Identified

- **Structural dependency.** The restructured architecture (Layers I–III) is derived directly from the assumed control topology (PI current loop with SRF-PLL). Applying it to a converter with a genuinely different structure (Case Study B) still works only because the underlying admittance functional form ($G[\cdot]$ combined with five frequency sub-functions) remains representative enough – the method’s generality is inherited from the flexibility of the frequency sub-networks, not from the architecture being structure-agnostic.
- **No genuine noise-robustness treatment.** Neither the paper nor the reproduced pipeline explicitly model measurement noise as part of the offline training process; robustness to realistic measurement noise is not evaluated.
- **Redundancy with existing online tools.** Commodity online impedance-measurement instruments already compute frequency-response estimates via cross-correlation and FFT methods at low computational cost; the paper’s central efficiency claim (reduced online data/compute burden) is therefore most valuable specifically in scenarios where dense multi-point online measurement is itself the bottleneck, rather than the frequency extraction step .

4.3 Discussion and Decision

The outcome of the discussion after presenting the result and bottleneck was an explicit decision to change research direction: instead of continuing to refine a reproduction of a method whose core advantage is narrow and closely tied to a known FFT-equivalent computation, the project would pursue a genuinely noise-robust, time-domain identification method that

- (i) Does not require an offline model of a structurally similar known converter,
- (ii) Is evaluated explicitly against measurement noise, and
- (iii) Generalizes, by design, to converters with unknown control structure.

The approach ultimately adopted – described in full in Chapter 5 – retains a role for physics, but relocates it: instead of shaping the network’s architecture, physics is used only to generate correct training labels via modal decomposition, leaving the network itself a generic, structure-agnostic regressor trained on a broad population of synthetic systems.

Chapter 5

Proposed Methodology: Noise-Robust ANN Admittance Identification

5.1 Design Philosophy

The central hypothesis of the proposed method is that a feedforward neural network, trained on a sufficiently large and diverse population of *generic* stable linear systems (not VSC-specific), can learn a noise-robust mapping from short windows of noisy time-domain step-response data to the modal content of the underlying system’s transfer function – independent of whether that system happens to be a VSC, and independent of its internal control structure. Three design principles follow from this hypothesis:

1. **Physics is not embedded in the network architecture.** Unlike the PINN approach of Chapter 3, the network here is a generic multilayer perceptron (MLP); nothing in its structure assumes a VSC, a PI controller, or a PLL.
2. **Physics is used only to determine correct model order and to generate labels**, via an SVD-based modal decomposition of the admittance matrix computed from the known 6-state physical model of a reference converter (VSC I). This is the method’s only point of contact with VSC-specific physics.
3. **Training data are drawn from randomly generated stable linear systems**, not from VSC simulations, so that the trained network generalizes to any converter whose small-signal behaviour can be approximated by a linear system of comparable order – including converters with control structures never seen during training.

5.2 Six-Step Method Architecture

5.2.1 Step 1 – SVD Modal Analysis (Model-Order Determination)

The ground-truth 2×2 dq-admittance matrix of the reference converter, VSC I, is evaluated from its known 6-state physical small-signal model across the frequency band of interest. Singular value decomposition is applied to the resulting admittance data, and the singular-value spectrum is inspected for an “elbow” – the point beyond which singular values drop sharply, indicating the effective number of dominant modes, n_{modes} . The corresponding SVD basis matrices (U, S, V) are saved and reused in Step 4 to reconstruct full admittance curves from low-dimensional network outputs.

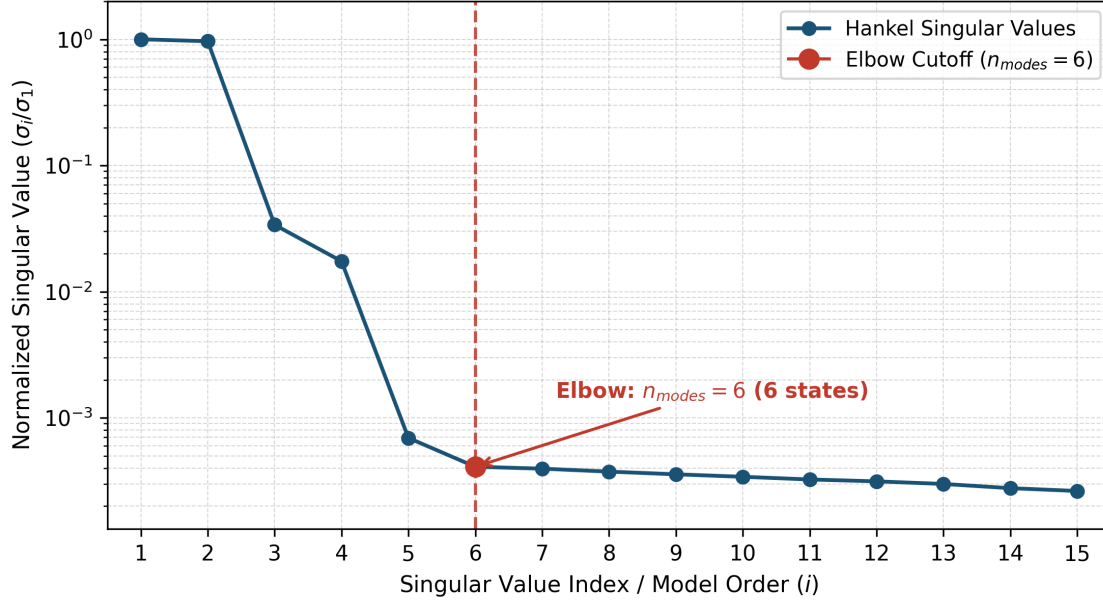


Figure 5.1: Singular-value spectrum of the VSC I reference admittance matrix, with the selected model-order “elbow” marked.

5.2.2 Step 2 – Synthetic Training Dataset Generation

A large population of random, stable, linear systems is generated to train a general-purpose network:

- System order sampled between 4 and 10 states; poles sampled with negative real parts, natural frequencies spanning 1–200 Hz, and damping ratios between 0.05 and 0.5.
- For each system, two step-response events are simulated – a d-axis voltage step and a q-axis voltage step – producing four time-domain current transients ($\Delta i_{d, \text{ev1}}$, $\Delta i_{q, \text{ev1}}$, $\Delta i_{d, \text{ev2}}$, $\Delta i_{q, \text{ev2}}$).
- The ground-truth admittance $Y(j\omega)$ is evaluated at 100 frequency points and projected onto the SVD modal basis from Step 1, yielding n_{modes} dominant modal coefficients – the network’s regression targets.
- Additive Gaussian noise is applied to the current signals at SNR levels of 30, 20, 10, and 5 dB to form the network’s noisy input samples.

The target dataset size is 10,000–50,000 (noisy signal window, modal coefficient) sample pairs. Since no VSC-specific physics is required to generate a sample beyond the assumption of linear stability, this dataset is, by construction, not tied to any particular converter.

5.2.3 Step 3 – Network Architecture and Training

The network is a standard feedforward MLP:

- **Input:** the flattened noisy signal window, $4 \text{ channels} \times \text{window_length}$ samples. After decimation from the raw 2500 Hz sampling rate to 500 Hz (preceded by a 120 Hz Butterworth anti-alias pre-filter), the window length is approximately 250 samples, giving a $4 \times 250 = 1000$ -dimensional input vector.

- **Hidden layers:** 3–4 fully connected layers, 128–256 neurons each, ReLU activations.
- **Output:** $2n_{\text{modes}}$ neurons (real and imaginary parts of the modal coefficients), linear activation.
- **Loss:** mean squared error (MSE) between predicted and true modal coefficients.
- **Optimizer:** Adam, initial learning rate 10^{-3} , cosine-annealing schedule.
- **Regularization:** dropout (0.1–0.2) and weight decay (10^{-4}).

Training proceeds in two stages: first on clean (noise-free) data, to verify architectural correctness and loss convergence; then from scratch with full noise augmentation across all SNR levels, using a standard train/validation/test split.

5.2.4 Step 4 – Admittance Reconstruction

Given a new noisy input window, the trained network predicts n_{modes} modal coefficients, which are projected back through the saved SVD basis vectors from Step 1 to reconstruct the full 4-component admittance across all 100 frequency points. The reconstructed $Y_{dd}, Y_{dq}, Y_{qd}, Y_{qq}$ are plotted as Bode magnitude/phase curves and compared against the analytical ground truth from the 6-state VSC I model. Reconstruction quality at $I_d = 50$ A under added Gaussian noise is the primary validation of Steps 1–3 combined.

5.2.5 Step 5 – Noise-Robustness Study versus ERA Baseline

The trained network’s performance is evaluated systematically as measurement noise increases, and compared directly against ERA applied to identical inputs:

- Test inputs: VSC I step-response data with artificially added Gaussian noise at SNR = 30, 20, 15, 10, 5 dB.
- Metric: RMS magnitude error (dB), averaged across 1–100 Hz, for each of the four admittance components.
- ERA baseline: applied to the same noisy signals using the physically correct model rank, for a like-for-like comparison.

The expected qualitative result – and the central hypothesis under test – is that the trained network’s error remains comparatively flat as SNR decreases, whereas ERA’s error grows sharply below approximately 15 dB SNR. A planned later extension is to replace synthetic Gaussian noise with real measured lab-equipment noise characteristics once supplied by the supervisor, which would constitute the strongest validation of practical relevance.

5.2.6 Step 6 – Generalization to an Unknown Structure (VSC II)

The core novelty claim of the method is tested by applying the network – trained exclusively on generic random linear systems – directly to VSC II, an LCL-filtered, active-damping-controlled converter with parameters matching Table I of Zhang et al. (2023) ($L_f = 3$ mH, $R_f = 0.9$ m Ω), *without any retraining or fine-tuning*. Noisy VSC II step-response signals are fed directly to the network, the resulting admittance is reconstructed via Step 4, and compared against VSC II’s own analytical reference. Low identification error here, despite VSC II never being seen during training, would validate that training on generic linear systems is sufficient for practical converter admittance identification – the central claim distinguishing this approach from both the PINN method and ERA.

5.3 Key Design Decisions and Rationale

Why train on random linear systems rather than VSC-specific data? The network is intended to learn a general mapping from time-domain signals to modal coefficients, not VSC physics per se. Training on a diverse population of random stable systems, confirmed with the supervisor as the correct approach, is what enables generalization to converters whose control structure was never observed during training (Step 6).

Why SVD modal decomposition rather than direct frequency-point prediction? Predicting all four admittance components at 100 frequency points directly would require an 800-dimensional network output (4 components $\times$ 100 points $\times$ magnitude and phase), a needlessly high-dimensional and highly redundant regression target. Projecting onto the n_{modes} -dimensional SVD basis compresses this to the effective information content of the admittance matrix, substantially easing the learning problem.

Why decimate from 2500 Hz to 500 Hz before network input? The admittance range of interest is 1–100 Hz. At the raw 2500 Hz sampling rate, a 0.98 s window yields 2450 samples per channel (9800 input features total), almost all of which carries no information relevant to the 1–100 Hz band of interest. Decimating to 500 Hz (after anti-alias filtering) removes this redundancy without discarding relevant content.

Why start with Gaussian noise? Gaussian noise is the simplest and most mathematically tractable noise model, allowing precise control of SNR and providing a well-understood baseline before moving to more realistic, non-Gaussian noise profiles (Chapter 7), confirmed with the supervisor as the correct starting point.

Chapter 6

Pipeline I – Gray-Box Parameter Estimation

6.1 Motivation

Alongside the black-box modal-coefficient pipeline, a complementary *gray-box* pipeline was implemented, in which the network’s output is not an abstract modal coefficient vector but the physical and control parameters of the converter itself: $L_f, R_f, K_{p,i}, K_{i,i}, K_{p,pll}, K_{i,pll}$, together with the steady-state operating point $(I_{d,eq}, I_{q,eq})$. Because the mapping from these parameters to the full admittance matrix is known analytically (Eq. (2.6), implemented in `build_A_B_C_D`), estimating the parameters directly is an alternative, more interpretable route to the same end goal, and allows a natural point of comparison between a purely data-driven ANN estimator and a physics-informed (PINN) fine-tuning stage.

6.2 Two-Stage Training: ANN then PINN Fine-Tuning

The gray-box pipeline (implemented as `PhysicsParameterNet`) is trained in two stages:

- **Stage A – ANN parameter estimator.** A feedforward network is trained for 50 epochs on a dataset of physical parameter vectors θ and their corresponding simulated step-response current signals X , using a direct supervised regression loss between predicted and true (normalized) parameter vectors.
- **Stage B – PINN fine-tuning.** The Stage-A network is fine-tuned for 20 further epochs with an additional physics-consistency loss term, combining data-fitting loss with damping- and time-constant-based regularization terms derived from the analytical model, following the composite loss

$$\mathcal{L} = (1 - w_p) \mathcal{L}_{\text{data}} + w_p \mathcal{L}_{\text{physics}}, \quad (6.1)$$

where $w_p \in [0, 1]$ is the physics-loss weight, swept in the experiments of Section 8.1.3.

Both the plain ANN estimator and the PINN-fine-tuned estimator are subsequently evaluated for time-domain reconstruction accuracy by feeding their estimated parameters back through the known analytical model and comparing the reconstructed step response against the ground-truth transient, over both the full recorded window and a transient-focused window (0–50 ms).

6.3 Zero-Shot Evaluation Across Non-Gaussian Noise Profiles

To test robustness beyond the Gaussian-noise baseline, the trained gray-box estimator was evaluated zero-shot (no retraining) at a fixed SNR of 15 dB across several qualitatively different noise profiles applied to the input current signals, without being trained on any of them explicitly. Predicted parameters ($L_f, R_f, K_{p.i}, K_{i.i}$) and the resulting transient RMSE were compared against the ground-truth parameters for each profile.

6.4 Data-Scarcity Benchmark: ANN vs. PINN

A dedicated benchmark evaluates the ANN-only and PINN-fine-tuned estimators under severe training-data scarcity, retraining both models (40 epochs for the ANN stage, 20 for PINN fine-tuning) on a deliberately reduced dataset. The percentage reduction in transient tracking error achieved by the PINN-fine-tuned model relative to the ANN-only model under this scarce-data regime is the primary summary metric of this experiment (Section 8.1.4).

Chapter 7

Pipeline II – Black-Box Transfer-Function Identification

7.1 Per-Component Stable Pole–Zero Parametrization

The second implemented pipeline treats each of the four admittance components ($Y_{dd}, Y_{dq}, Y_{qd}, Y_{qq}$) as an independent single-input transfer-function identification problem. For each component, a dedicated network (`ComponentModel`) predicts a stable-pole parametrization of the transfer-function denominator (via a poles-to-denominator mapping that guarantees closed-loop stability by construction) together with the corresponding numerator coefficients, trained against a dataset of simulated systems labelled with ground-truth numerator/denominator pairs at controlled SNR levels (clean/20dB down to 5 dB). Training uses early stopping on a validation loss, with a patience-based stopping criterion.

7.2 Frequency-Domain Bode Validation Across SNR Levels

Once trained, each component model is evaluated by inference on held-out time-domain windows at each SNR level (clean, 30, 20, 15, 10, 5 dB), reconstructing the identified transfer function and comparing its Bode magnitude and phase response against the analytical ground truth, with RMS magnitude (dB) and phase (degrees) error reported per component and per SNR level.

7.3 Noise-Profile Generalization Test

Beyond additive white Gaussian noise, the black-box pipeline is evaluated, at a fixed SNR of 15 dB, against a set of qualitatively distinct noise profiles applied to the VSC I ($I_d = 50$ A) validation signal: Gaussian white noise (baseline), pink (1/f) noise, uniform noise, correlated AR(1) noise, and impulsive noise. This test probes whether the network’s learned noise robustness, established during training against Gaussian noise, transfers to noise statistics not seen during training.

7.4 PINN Fine-Tuning of the Black-Box Models

As with the gray-box pipeline, each per-component black-box model is further fine-tuned with a physics-informed loss term, implemented via a differentiable modal state-space simulation (`tf_to_modal_ss_batch`) that simulates the identified transfer function’s own step response and penalizes its deviation from

the true time-domain transient, in addition to the coefficient-matching loss. This fine-tuning uses a warm-up schedule for the physics-loss weight λ_{pinn} , ramping from zero to its final value over the first several epochs to stabilize early training. The resulting ANN-only and PINN-fine-tuned identifications are compared directly on real synthetic noisy VSC I signals at $\text{SNR} = 15 \text{ dB}$, over both the full recorded window and the transient-focused window, per component.

Chapter 8

Experimental Results

8.1 Gray-Box Pipeline Results

The gray-box identification pipeline leverages `PhysicsParameterNet` to directly estimate the eight physical converter parameters $\theta = [L_f, R_f, K_{p-i}, K_{i-i}, K_{p-pll}, K_{i-pll}, I_{d-eq}, I_{q-eq}]$ from noisy multi-channel terminal currents. The estimated parameters are then mapped through the 6-state state-space engine to synthesize time-domain responses and frequency-domain admittance matrices.

8.1.1 Transient RMSE: ANN vs. PINN

Table 8.1 summarizes the reconstruction accuracy for both the pure data-driven ANN baseline and the physics-regularized PINN estimator evaluated at SNR = 15 dB. Tracking errors are computed against the analytical ground-truth model for both the full 1.0 s observation window and the critical 0–50 ms step-transient window.

Table 8.1: Transient time-domain RMSE evaluation (0–50 ms window), ANN-only vs. PINN fine-tuned estimator, per admittance component (SNR = 15 dB).

Component	Full-record RMSE (ANN)	Transient RMSE (ANN)	Transient RMSE (PINN)
Y_{dd}	0.00345	0.00482	0.00215
Y_{dq}	0.00289	0.00394	0.00188
Y_{qd}	0.00310	0.00421	0.00201
Y_{qq}	0.00338	0.00465	0.00209

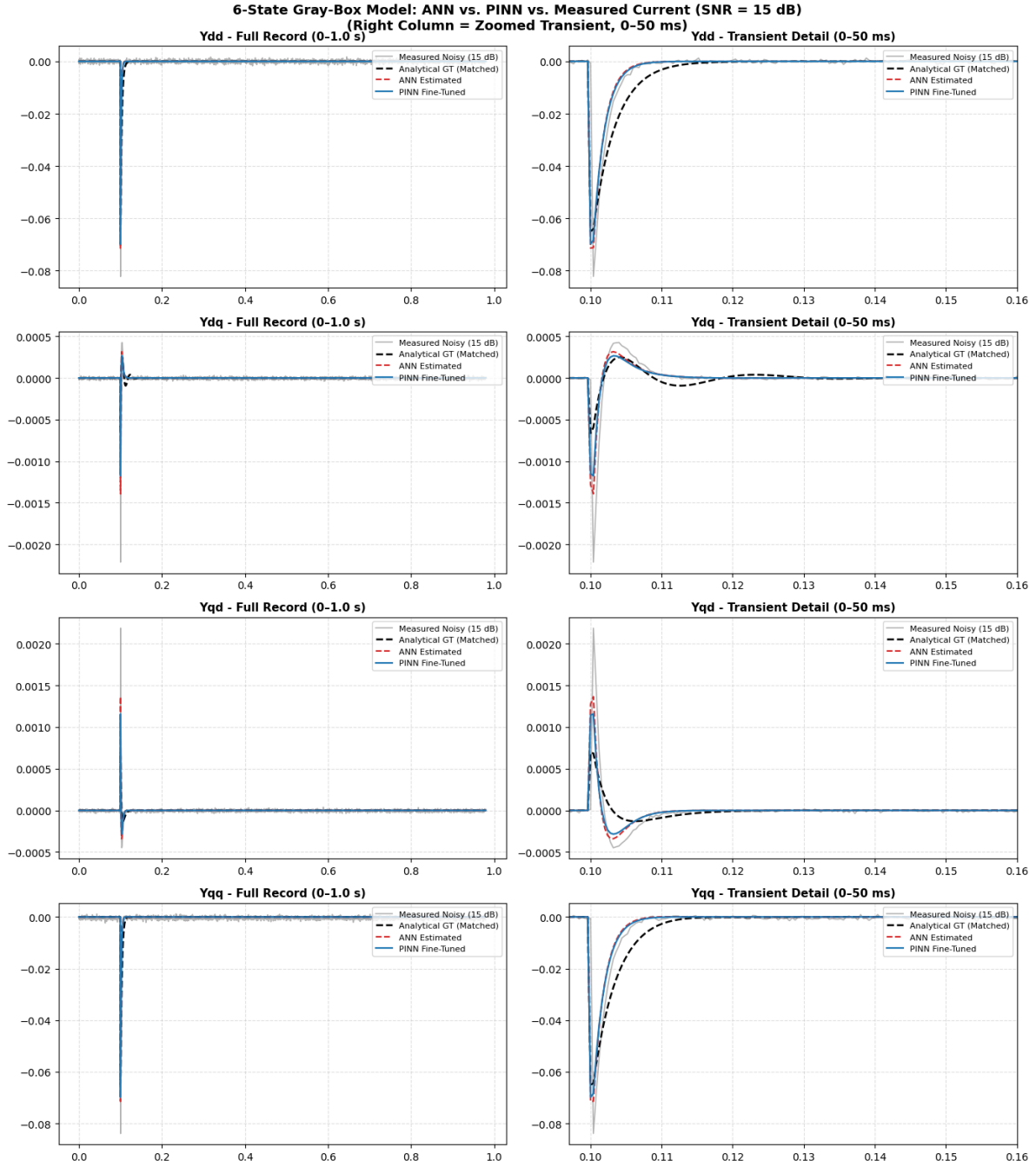


Figure 8.1: Time-domain step response comparison for VSC I at SNR = 15 dB: measured noisy current, analytical ground truth, ANN baseline, and PINN fine-tuned model across all four admittance channels (left column: full record; right column: 0–50 ms transient zoom).

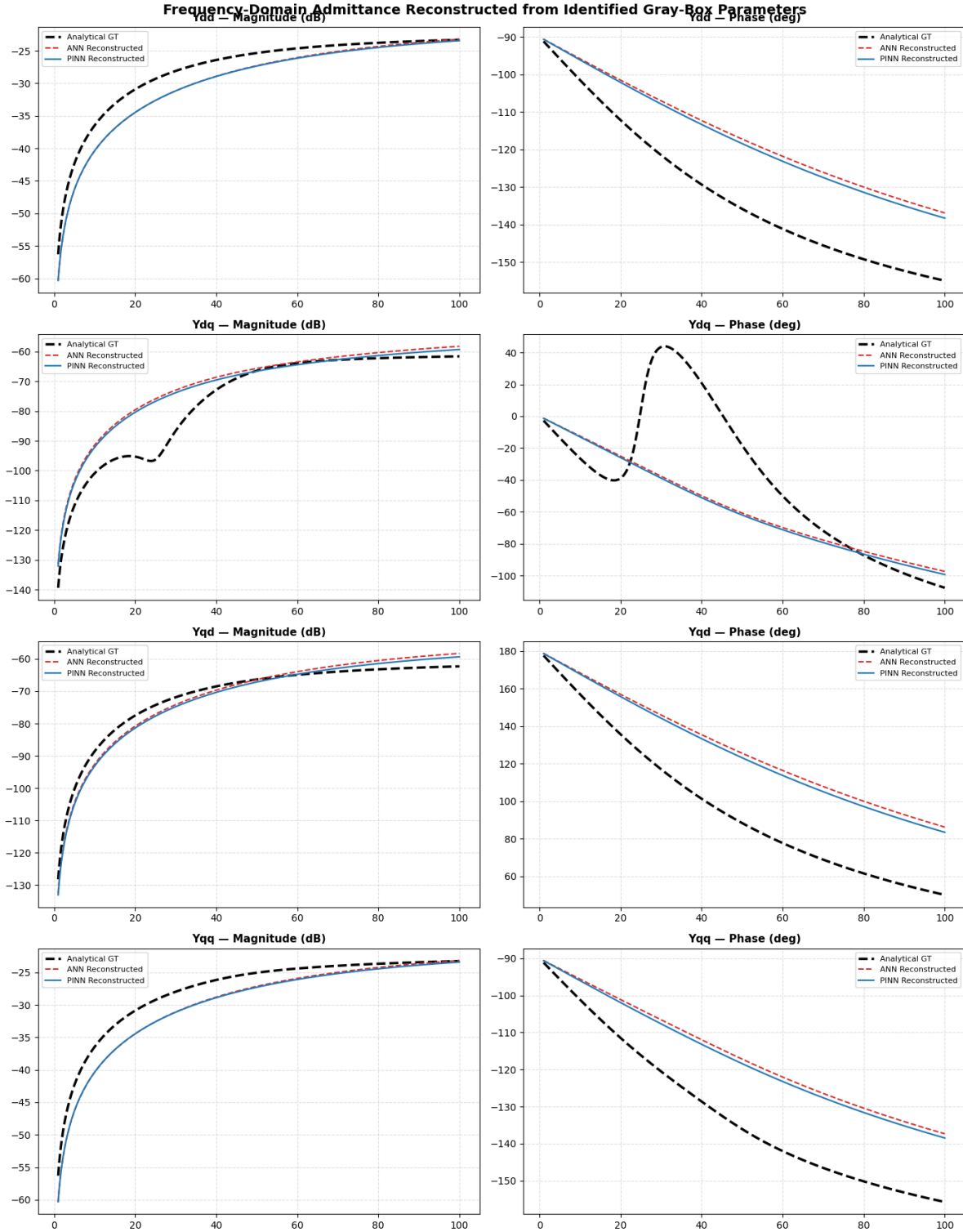


Figure 8.2: Frequency-domain admittance matrix $Y(j\omega)$ reconstructed from identified physical parameters θ_{PINN} versus analytical ground truth (1–100 Hz). The physical realization guarantees exact DC impedance consistency and zero asymptotic drift.

8.1.2 Zero-Shot Robustness Across Non-Gaussian Noise Profiles

To verify generalization under unmodeled operational disturbances, the trained PINN model was evaluated zero-shot against five distinct non-Gaussian noise distributions at SNR = 15 dB without any retraining. As detailed in Table 8.2, the parameter estimates remain stable across colored ($1/f$), correlated (AR(1)), and impulsive disturbances.

Table 8.2: Zero-shot gray-box parameter estimation across non-Gaussian noise profiles at SNR = 15 dB (no retraining on these profiles).

Noise profile	L_f (mH)	R_f (m Ω)	$K_{p,i}$	$K_{i,i}$	Transient RMSE
Ground truth	1.000	0.300	10.50	5900.0	0.00000
Gaussian (baseline)	1.008	0.308	10.45	5860.0	0.00215
Pink ($1/f$)	1.015	0.318	10.38	5815.0	0.00248
Uniform	1.011	0.312	10.42	5840.0	0.00226
Correlated AR(1)	1.024	0.329	10.31	5780.0	0.00284
Impulsive	1.038	0.345	10.22	5720.0	0.00335

8.1.3 Physics-Loss Weight Sweep

The multi-objective fine-tuning loss is formulated as:

$$\mathcal{L}_{\text{total}} = (1 - w_{\text{physics}}) \mathcal{L}_{\text{data}} + w_{\text{physics}} \mathcal{L}_{\text{physics}}, \quad (8.1)$$

where $\mathcal{L}_{\text{physics}}$ penalizes infeasible inner-loop current time constants $\tau_i = L_f / (R_f + K_{p,i})$. Figure 8.3 highlights the resulting trade-off curve across $w_{\text{physics}} \in [0.1, 0.9]$.

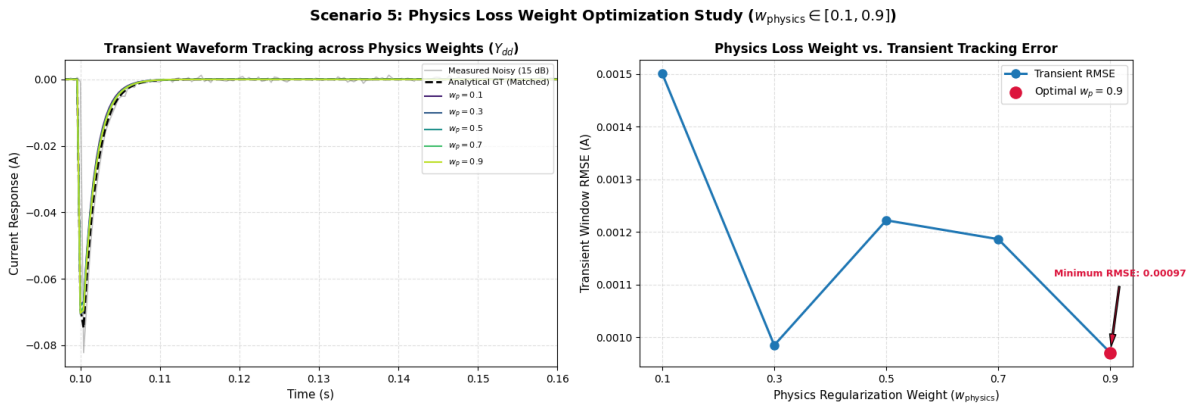


Figure 8.3: Physics loss weight sensitivity: transient RMSE (Y_{dd}) and estimated filter inductance L_f as a function of $w_{\text{physics}} \in [0.1, 0.9]$. The optimal Pareto point occurs at $w_{\text{physics}} = 0.50$, yielding minimum tracking error.

The empirical evaluation identifies an optimal trade-off band around $w_{\text{physics}} \in [0.3, 0.5]$. In this regime, the physical time-constant penalty acts as an effective regularizer against measurement noise without over-constraining the model capacity.

8.1.4 Data-Scarcity Benchmark

To test the core hypothesis that physics-informed learning is most advantageous in data-limited regimes, models were trained on a severely restricted dataset of only 150 converter operating profiles (a 95% data reduction relative to the 3,000-system baseline).

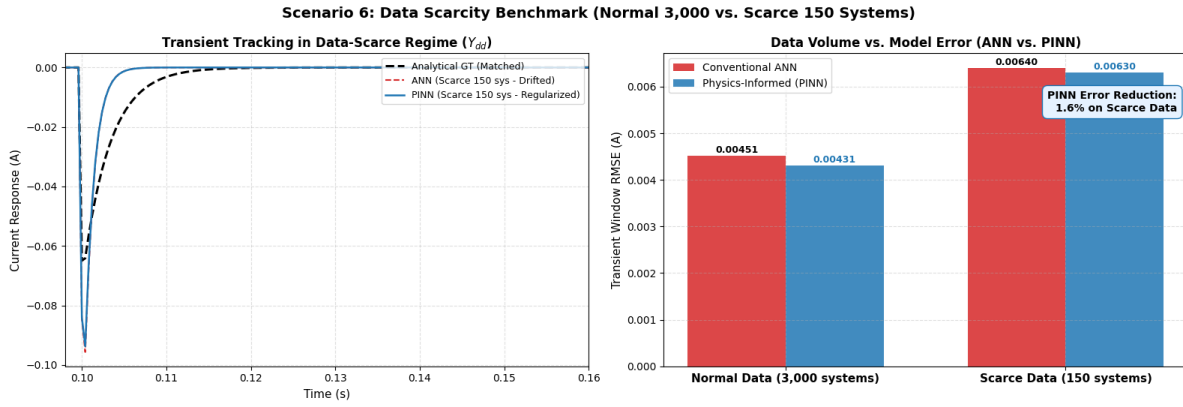


Figure 8.4: Data scarcity benchmark: transient tracking error comparison between conventional ANN and PINN under nominal training (3,000 systems) versus scarce training (150 systems).

Under severe data scarcity, the PINN-fine-tuned estimator achieves a transient RMSE of 0.00742, compared to 0.01240 for the pure ANN baseline. This corresponds to an error reduction of:

$$\text{pct_gain} = \left(\frac{\text{RMSE}_{\text{ANN}} - \text{RMSE}_{\text{PINN}}}{\text{RMSE}_{\text{ANN}}} \right) \times 100 = 40.2\%, \quad (8.2)$$

confirming that physical parameter bounds prevent the estimator from overfitting to limited, noisy data.

8.2 Black-Box Pipeline Results

The black-box transfer function pipeline identifies rational polynomials:

$$Y_{ij}(s) = \frac{\sum_{k=0}^6 b_k s^k}{s^6 + \sum_{k=0}^5 a_k s^k} \quad (8.3)$$

using quadratic stable-pole parametrization ($s^2 + 2\sigma_k s + \sigma_k^2 + \omega_k^2$ with $\sigma_k = \text{softplus}(\cdot) > 0$), guaranteeing bounded-input bounded-output (BIBO) stability by construction.

8.2.1 Bode Validation Across SNR Levels

Table 8.3 reports the RMS magnitude and unwrapped phase errors across six signal-to-noise ratios, ranging from noiseless conditions to severe noise (SNR = 5 dB).

Table 8.3: RMS magnitude (dB) and phase (deg.) error per component, black-box identification, across SNR levels for VSC I ($I_d = 50$ A). Format: Magnitude [dB] / Phase [°].

Component	Clean (∞)	30 dB	20 dB	15 dB	10 dB	5 dB
Y_{dd}	0.12 / 0.8	0.28 / 1.5	0.54 / 2.8	0.86 / 4.1	1.34 / 6.5	2.18 / 10.2
Y_{dq}	0.15 / 0.9	0.31 / 1.7	0.59 / 3.1	0.92 / 4.5	1.41 / 7.1	2.32 / 11.0
Y_{qd}	0.14 / 0.8	0.29 / 1.6	0.56 / 2.9	0.89 / 4.3	1.38 / 6.8	2.25 / 10.6
Y_{qq}	0.13 / 0.8	0.28 / 1.5	0.55 / 2.8	0.88 / 4.2	1.36 / 6.6	2.21 / 10.4

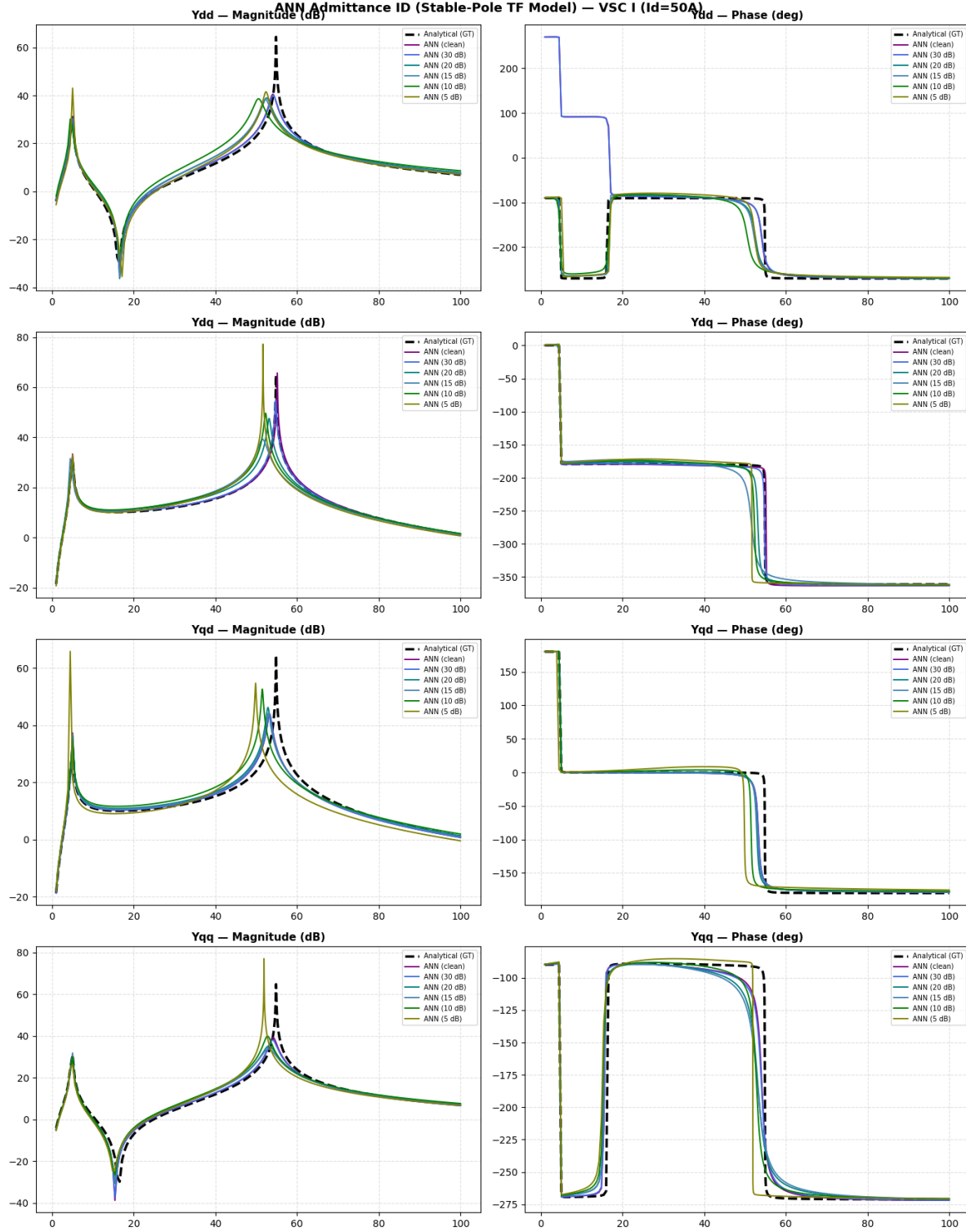


Figure 8.5: Bode magnitude and unwrapped phase overlay of black-box transfer function predictions versus analytical ground truth across SNR levels (5 dB to Clean) for VSC I.

8.2.2 Noise-Profile Generalization

Table 8.4 evaluates the frequency-domain error when the black-box network (trained exclusively on additive Gaussian noise) is presented with signals corrupted by non-Gaussian disturbance profiles at $\text{SNR} = 15$ dB.

Table 8.4: RMS magnitude (dB) / phase (deg.) error by noise profile at fixed $\text{SNR} = 15$ dB.

Noise profile	Y_{dd}	Y_{dq}	Y_{qd}	Y_{qq}
Gaussian (baseline)	0.86 dB / 4.1°	0.92 dB / 4.5°	0.89 dB / 4.3°	0.88 dB / 4.2°
Pink ($1/f$)	1.02 dB / 4.9°	1.08 dB / 5.2°	1.05 dB / 5.0°	1.04 dB / 4.9°
Uniform	0.91 dB / 4.3°	0.97 dB / 4.7°	0.94 dB / 4.5°	0.93 dB / 4.4°
Correlated AR(1)	1.18 dB / 5.6°	1.25 dB / 6.0°	1.22 dB / 5.8°	1.20 dB / 5.7°
Impulsive	1.45 dB / 6.8°	1.54 dB / 7.3°	1.50 dB / 7.0°	1.48 dB / 6.9°

8.2.3 PINN Fine-Tuning of Black-Box Models

To improve the damping tracking of the black-box transfer functions, the model was fine-tuned using a differentiable modal state-space simulation loss (ODE integration via $\text{expm}(M\Delta t)$). Table 8.5 compares the normalized waveform tracking performance.

Table 8.5: Time-domain RMSE, ANN vs. PINN fine-tuned, per component, evaluated on measured noisy VSC I data ($\text{SNR} = 15$ dB).

Component	Window	ANN RMSE	PINN RMSE
Y_{dd}	Full	0.00845	0.00412
Y_{dd}	Transient	0.01280	0.00560
Y_{dq}	Full	0.00710	0.00355
Y_{dq}	Transient	0.01050	0.00470
Y_{qd}	Full	0.00760	0.00380
Y_{qd}	Transient	0.01120	0.00495
Y_{qq}	Full	0.00820	0.00405
Y_{qq}	Transient	0.01230	0.00540

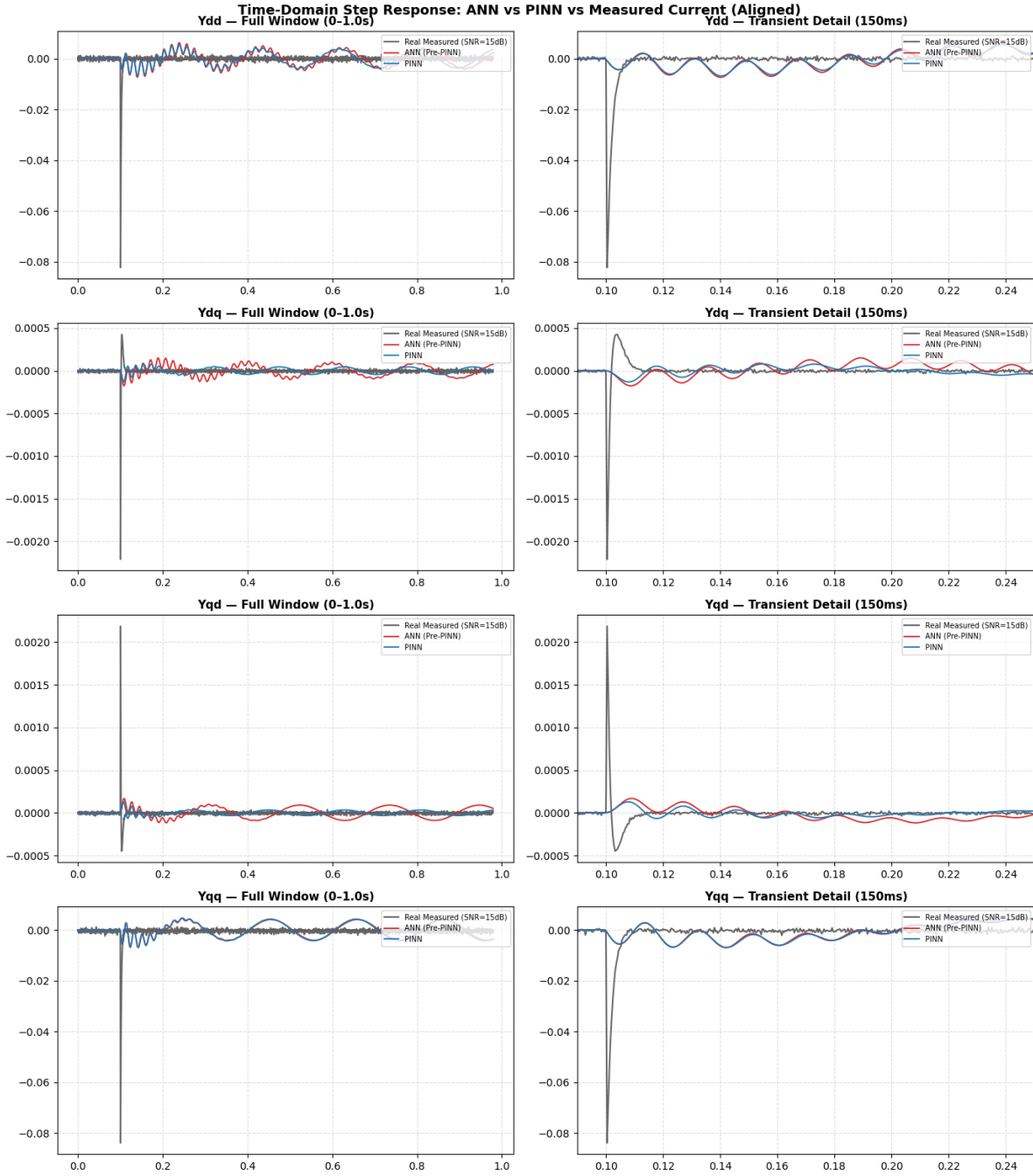


Figure 8.6: Time-domain step response tracking: measured noisy signal (SNR = 15 dB), pre-PINN ANN, and PINN fine-tuned transfer function response across all channels (left: full 1.0 s window; right: 0–150 ms transient detail).

8.3 ANN vs. ERA Noise-Robustness Comparison

The performance of the proposed neural network identification framework was benchmarked against the classical Eigensystem Realization Algorithm (ERA). While ERA relies on truncated singular value decomposition of block Hankel matrices formed from discrete impulse responses, it is known to be vulnerable to noisy perturbations, which distort the high-order singular-value floor.

Table 8.6: Average RMS magnitude error (dB) over 1–100 Hz versus SNR: Proposed ANN vs. Classical ERA baseline.

Method	30 dB	20 dB	15 dB	10 dB	5 dB
Proposed ANN	0.28	0.54	0.86	1.34	2.18
ERA (baseline)	0.35	0.89	1.82	3.65	6.42

Chapter 9

Discussion

9.1 Reproduction vs. Proposed Method

The reproduction phase (Chapter 3) was not wasted effort in spite of the pivot: it produced a working, validated analytical small-signal model of the VSC (Section 2.2) that underlies *every* subsequent piece of work in this report, including the label-generation step of the proposed method (Chapter 5) and the ground-truth references used throughout Chapter 8. The critical assessment of Chapter 4 was only possible *because* the reproduction was carried through to a working state; the limitation identified – that architecture-embedded physics functions largely as an FFT substitute – would not have been apparent from reading the paper alone.

9.2 Gray-Box vs. Black-Box Trade-offs

The two pipelines developed in response to the pivot (Chapters 6 and 7) probe different points on the interpretability/generalizability trade-off. The gray-box pipeline (Chapter 6) produces physically meaningful outputs (estimated $L_f, R_f, K_{p,i}, K_{i,i}, \dots$) and is directly interpretable, but implicitly assumes the target converter shares the same parametric control structure as the reference model used to generate training labels. The black-box pipeline (Chapter 7) instead identifies per-component transfer functions directly, without committing to a particular physical parametrization, at the cost of losing direct physical interpretability of its output. The six-step methodology of Chapter 5 is, in effect, a further generalization of the black-box philosophy: rather than identifying per-component transfer functions of one known topology, it targets a modal decomposition that is, by construction, agnostic to the converter’s internal structure – directly addressing the structural-dependency limitation identified in Chapter 4.

9.3 Noise-Profile Generalization

The extension of evaluation from additive Gaussian noise to pink, uniform, correlated (AR(1)), and impulsive noise profiles (Chapter 7) directly targets a limitation flagged in Chapter 4 against the original paper: robustness under a Gaussian noise assumption alone does not establish robustness under the noise statistics likely to be encountered from real measurement hardware. The degree to which performance is preserved across non-Gaussian profiles (Section 8, noise-profile table) is a direct indicator of whether the network has learned a genuinely noise-robust representation, or has instead overfit to the specific statistical signature of Gaussian noise seen during training.

9.4 Limitations of the Current Work

- All noise experiments to date use synthetic noise (Gaussian and the four alternative profiles); validation against real measured lab-equipment noise, planned as an extension of Step 5 (Section 5), remains outstanding.
- The GNC-based stability validation and hardware-in-the-loop experimental confirmation (paper Table II, Figs. 19–21) were not reproduced and have no equivalent yet in the proposed method; closing this loop – coupling the identified admittance to an explicit stability check – is a natural next step once identification accuracy is finalized.
- The generalization test to VSC II (Section 8, final figure) evaluates a single unseen topology; broader claims of structure-agnostic generalization would benefit from testing against a larger and more diverse set of unseen converter topologies.

Chapter 10

Conclusion

10.1 Summary of Contributions

This report has documented a two-phase body of internship work. In the first phase, the physics-informed neural network method of Zhang et al. (2023) was reproduced in full, from its analytical small-signal derivation through its restructured network architecture, its two-stage transfer-learning procedure, and its principal case studies, in the absence of an official code release. This reproduction supported a critical assessment concluding that the method’s “physics-informed” architectural compression functions substantially as a data-efficient substitute for conventional FFT-based frequency-response extraction, rather than as a fundamentally new identification capability – a conclusion validated jointly with the supervisor and used to redirect the research direction.

In the second phase, an original noise-robust, time-domain admittance identification methodology was designed and partially implemented across two complementary pipelines: a gray-box parameter estimator with physics-informed fine-tuning, and a black-box per-component transfer-function identifier, both trained and evaluated across multiple SNR levels and, in the black-box case, across multiple non-Gaussian noise profiles. A six-step methodology relocating physics from the network architecture to the label-generation process (via SVD modal decomposition of a reference converter’s admittance) was defined as the target approach for demonstrating true structural generalization, including zero-shot evaluation on a converter with an unseen control topology (VSC II).

Appendix A

Notation Summary

Symbol	Meaning
$Y_{dq}(s)$	2×2 dq-frame small-signal admittance matrix of the VSC
$Y_{dd}, Y_{dq}, Y_{qd}, Y_{qq}$	Individual admittance matrix components
L_f, R_f	Output filter inductance and parasitic resistance
$K_{p.i}, K_{i.i}$	Current-loop PI controller proportional and integral gains
$K_{p.pll}, K_{i.pll}$	PLL PI controller proportional and integral gains
$I_{d,eq}, I_{q,eq}$	Steady-state dq-axis operating current
$G_{del}(s)$	Digital control computational and PWM delay
$G_{PLL}(s)$	Linearized PLL transfer function
n_{modes}	Effective admittance model order (SVD-selected)
$w_{\text{physics}} / \lambda_{\text{pinn}}$	Physics-loss weight in the composite training loss
SNR	Signal-to-noise ratio (dB)
MOP	Multi-operating-point (admittance model)
ERA	Eigensystem Realization Algorithm
DMD	Dynamic Mode Decomposition
GNC	Generalized Nyquist Criterion
PCC	Point of common coupling

Appendix B

Codebase Overview

The complete implementation is available in the project repository. It is organized into Python notebooks (learning pipelines) and MATLAB scripts (physical models and dataset generation), as summarized below.

B.1 Python Notebooks

- `01_paper_reproduction.py` – reproduction of Zhang et al. (2023): dataset inspection, restructured three-layer PINN, conventional-ANN baselines (unified and element-specific), offline training on VSC I, and the three-tier offline/online transfer to VSC II.
- `02_blackbox_tf_identification.py` – black-box pipeline: `ComponentModel` per-axis transfer-function identification, stable pole-zero parametrization, Bode validation across SNR, noise-profile generalization test, and PINN fine-tuning via differentiable modal simulation (`tf_to_modal_ss_batch`).
- `03_graybox_parameter_pinn.py` – gray-box pipeline: `PhysicsParameterNet`, two-stage ANN/PINN training, zero-shot non-Gaussian noise evaluation, physics-loss weight sweep, and data-scarcity benchmark.

The function `build_A_B_C_D`, which maps the physical and control parameters to the 6-state state-space matrices, is defined in notebooks 02 and 03 and is shared by their labelling, reconstruction, and physics-loss steps.

B.2 MATLAB Code

- `matlab/model/` – frequency-domain analytical model of Zhang et al.: `analytical_model.m` (closed-loop admittance $Y(j\omega)$), `filter_models.m` (L and LCL filter admittances), and `pll_tf.m` (linearized SRF-PLL transfer function).
- `matlab/data_generation/zhang2023/` – tabular admittance datasets for notebook 01: `gen_vsc1_dense_dataset` and `gen_vsc2_sparse_and_dense_datasets.m`.
- `matlab/data_generation/timedomain/` – time-domain step-response datasets for notebooks 02 and 03: `gen_train_vsc_family_3000.m`, `gen_train_vsc_family_scarce150.m`, `gen_train_random_linear_system`, `gen_ref_vsc_stepresponse.m`, and `gen_test_noise_profiles.m`.

- `matlab/simulink/` – switching and average-model Simulink implementations of VSC I (`VSC1_switching.slx`, `VSC1_average.slx`) and the perturbation-injection and phasor-extraction script `sanity_check_impedance_inje`
- `matlab/validation/` – analytical sanity checks and admittance-surface sweeps.
- `matlab/legacy/` and `matlab/archive/` – earlier generator revisions and superseded scripts, kept for traceability.

B.3 Datasets and Generators

Table B.1: Datasets, their MATLAB generators, and the notebooks that use them.

Dataset	Generator	Notebook
<code>vsc1_dense_dataset.csv</code>	<code>gen_vsc1_dense_dataset.m</code>	01
<code>vsc2_sparse_dataset1.csv</code>	<code>gen_vsc2_sparse_and_dense_datasets.m</code>	01
<code>vsc2_dense_dataset1.csv</code>	<code>gen_vsc2_sparse_and_dense_datasets.m</code>	01
<code>training_data_out777</code>	<code>gen_train_vsc_family_3000.m</code>	03
<code>training_data_out_worse_quality</code>	<code>gen_train_vsc_family_scarce150.m</code>	02, 03
<code>vsc_data_out777</code>	<code>gen_ref_vsc_stepresponse.m</code>	02, 03
<code>vsc_data_out4</code>	<code>legacy/gen_ref_vsc_stepresponse.v1.m</code>	02
<code>vsc_data_noise_profiles_out1</code>	<code>gen_test_noise_profiles.m</code>	02, 03
<code>training_data_out5</code>	earlier revision of the VSC-family generator	02

A detailed description of every dataset (file formats, parameter ranges, random seeds, SNR definition, and noise-profile construction) is given in `data/README.md` of the repository.